

Learning Objectives

Chapter 1: The Nature of Science & Physics

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1. Define physics as the study of energy, matter, space, time, and their interactions; understand the role of scientific laws and theories in describing nature.

2. Identify the SI base units used in this text (length, mass, time, current) and their symbols.

meter (m), kilogram (kg), second (s), ampere (A)

3. Use metric prefixes and scientific notation to express quantities over many orders of magnitude; apply prefix conversions and recognize common examples (kilo, milli, micro, nano, etc.). See the compact reference table below for prefixes, symbols, powers of ten, and example uses.

Prefix	Symbol	Value	Example (typical)
tera	T	10^{12}	terawatt (TW) $\sim 10^{12}$ W — powerful laser output
giga	G	10^9	gigahertz (GHz) $\sim 10^9$ Hz — microwave frequency
mega	M	10^6	megacurie (MCi) $\sim 10^6$ Ci — high radioactivity
kilo	k	10^3	kilometer (km) = 10^3 m — about 0.6 mile
—	—	10^0	(unity)
deci	d	10^{-1}	deciliter (dL) = 10^{-1} L — less than half a soda
centi	c	10^{-2}	centimeter (cm) = 10^{-2} m — fingertip thickness
milli	m	10^{-3}	millimeter (mm) = 10^{-3} m — flea height at shoulders
micro	μ	10^{-6}	micrometer (μ m) = 10^{-6} m — microscope detail
nano	n	10^{-9}	nanogram (ng) = 10^{-9} g — tiny dust speck
pico	p	10^{-12}	picofarad (pF) = 10^{-12} F — small radio capacitor
femto	f	10^{-15}	femtometer (fm) = 10^{-15} m — proton size scale

4. Perform unit conversions by using conversion factors.

Example 1: Length Conversion

Convert 5.0 km to meters. (the above table shows that the prefix *kilo* represents a value of 10^3 $\therefore$ 1 km = 10^3 m)

$$5.0 \text{ km} \times \frac{10^3 \text{ m}}{1 \text{ km}} = 5000 \text{ m}$$

Example 2: Time Conversion

Convert 2.5 hr to seconds.

$$2.5 \text{ hr} \times \frac{60 \text{ min}}{1 \text{ hr}} \times \frac{60 \text{ s}}{1 \text{ min}} = 9000 \text{ s}$$

5. Express measured values with uncertainty using the standard $\pm$ notation and propagate the idea of a measured result.

$$x = x_0 \pm \Delta x$$

6. Distinguish accuracy (closeness to true value) from precision (repeatability) and use appropriate reporting: round final answers to reflect measurement precision (example formatting).

$$3.14159 \approx 3.14$$

7. Use approximation and order-of-magnitude estimates to simplify problem solving; practice writing and manipulating powers of ten for back-of-envelope checks.